

# **AC7923B Datasheet**

**Zhuhai Jieli Technology Co.,LTD**

**Version 1.4**

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## Revision History

| Date       | Revision | Author   | Description                                                                    |
|------------|----------|----------|--------------------------------------------------------------------------------|
| 2024.05.15 | V1.0     | zh-jieli | Initial Release                                                                |
| 2024.05.30 | V1.1     | zh-jieli | Pin Adjustment                                                                 |
| 2024.09.30 | V1.2     | zh-jieli | Add Power Domain Information                                                   |
| 2024.12.07 | V1.3     | zh-jieli | Modify Audio DAC Characteristics, IO Characteristics and Operating Temperature |
| 2025.01.09 | V1.4     | zh-jieli | Update Feature_Bluetooth, Update Block Diagram                                 |
|            |          |          |                                                                                |

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# AC7923B Features

## SYSTEM

- Dual Core 32bit DSP 320MHz
- With IEEE754 Single precision FPU
- Support jieli TEE
- Support FFT / MATRIX / MATH
- 2 x I-cache and D-cache
- On-chip SRAM 352kbyte
- Support SDTAP / EMU / ETM
- Support MMU
- Support MPU
- Built-in SDRAM/DDR (Maximum 64Mbyte)
- SPI FLASH Controller (Maximum 64Mbyte)
- 24MHz crystal oscillator
- 32KHz RTC crystal oscillator
- Internal RC oscillator, PLL

## Video Input

- Internal Image Signal Processor
- Support DVP, BT656, SPI interface
- Support 1 lane MIPI-CSI interface
- Support RAW, YUV422 formats
- Support video resize an time mark
- 2 x JPEG codec

## Video output

- Support display color enhancement
- Support DPI, DBI, BT656 interface
- Support RGB, YUV formats

## Graphics

- Internal 2D DMA
- Internal 2.5D GPU
- Support vector graphics rendering
- Support image resize, rotation, projection
- Support multiple blending mode
- Support ARGB, RGB, YUV, Lx, Ax formats

## DSP Audio Processing

- SBC/AAC/LDAC/LHDC/LC3/CVSD/mSBC codec
- mSBC voice codec supported for BT phone
- PLC for voice processing

- Single/Multi MIC ENC
- Multi-band DRC
- Multi-band EQ
- Support spatial sound

## Audio

- 2 x 16bit DAC
  - ❖ SNR 103dB
  - ❖ Noise 6.4uVrms
  - ❖ Supports differential mode
  - ❖ Sampling rate 8~96kHz
- 2 x 16bit ADC
  - ❖ SNR 95dB
  - ❖ Sampling rate 8~48kHz
- I2S/PDM AUDIO Master/Slave interface

## Bluetooth

- Dual-mode BT6.0 with LE Audio (DN Q332415)
- Support AoA/AoD
- Support LE audio BIS/CIS
- Support long range BLE
- Maximum transmitting power 19 dBm
- Receiver sensitivity
  - ❖ -95.5 dBm @BR
  - ❖ -96 dBm @EDR  $\pi/4$  DQPSK
  - ❖ -88 dBm @EDR 8DPSK

## IEEE 802.11b/g/n

- 1T1R in 2.4 GHz band
- 20 MHz and 40 MHz bandwidth
- Data rate up to 150 Mbps
- Security:WFA/WPA3 personal,WPS2.0,WAPI
- QoS: WFA WMM, WMM PS
- Support STBC, A-MPDU, A-MSDU, BLK-ACK
- Support Station, SoftAP,Station+SoftAP, Promiscuous mode
- Maximum transmitting power
  - ❖ 19 dBm @1Mbps, DSSS
  - ❖ 17 dBm @HT20, MCS0
  - ❖ 13 dBm @HT20, MCS7

- Receiver sensitivity
  - ❖ -97 dBm @1Mbps, DSSS
  - ❖ -93 dBm @HT20, MCS0
  - ❖ -74 dBm @HT20, MCS7

### Peripherals

- 1 x High speed USB
- 2 x SD host controller
- 6 x Multi-function 32bit timer
- 5 x UART interface
- 3 x I<sup>2</sup>C Master/Slave interface
- 3 x SPI Master/Slave interface
- 1 x QDEC
- 1 x CAN Controller
- 8 x MCPWM
- 1 x PAP Interface
- 3 x Light strip Controller
- 1 x 10bit ADC(4 Channel)
- 43 x GPIO Support function remapping
- Built-in RTC with alarm, wakeup

### PMU

- 1 x Buck DC-DC converter
- 2 x IO power domain
- WIFI PA support external power supply
- RTCVDD33 support external power supply
- Support temperature sensor
- VBAT range 2.7V to 5.5V
- IOVDD range 2.7V to 3.6V

### Packages

- QFN80(8mm\*8mm)

### Temperature

- Operating temperature
  - TC = -20℃ to +85℃(standard range)
- Storage temperature -65℃ to +150℃

### Applications

- IPC
- Driving recorder
- WIFI Appliances

## 1 Block Diagram

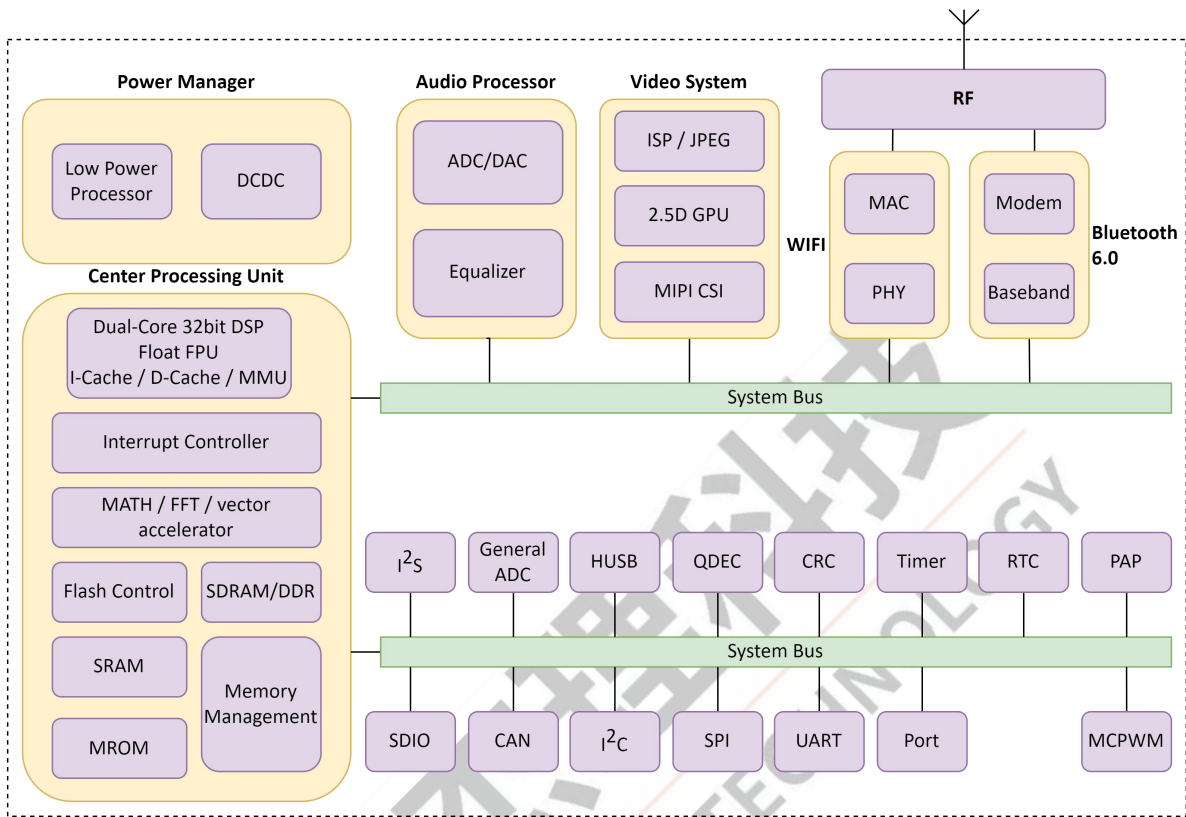


Figure 1-1 AC7923B Block Diagram

## 2 Pin Definition

### 2.1 Pin Assignment

|               | ANTP | WFVDD33 | WFVDD14 | WFVSS | PA1 | PA2       | PA3 | PA4 | PA5 | PA6 | PA7 | PA8     | PA9 | PA10 | PA11  | AVD12 | CS1 D0N | CS1 D0P | CS1 DIN | CS1 DIP |             |
|---------------|------|---------|---------|-------|-----|-----------|-----|-----|-----|-----|-----|---------|-----|------|-------|-------|---------|---------|---------|---------|-------------|
| EVSS          | 80   | 79      | 78      | 77    | 76  | 75        | 74  | 73  | 72  | 71  | 70  | 69      | 68  | 67   | 66    | 65    | 64      | 63      | 62      | 61      |             |
| WFVDD14       | 1    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | IOVDD/AVD33 |
| XOSCI         | 2    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | DVDD        |
| XOSCO         | 3    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PA12        |
| PE13          | 4    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PA13        |
| PE12          | 5    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PA14        |
| PE11          | 6    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PA15        |
| PE10          | 7    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB0         |
| IOVDD2        | 8    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB1         |
| PE0           | 9    |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB2         |
| PD15          | 10   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB3         |
| PD14          | 11   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB4         |
| PD13          | 12   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB5         |
| PD12          | 13   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB6         |
| PD11          | 14   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PB7         |
| PD10          | 15   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PC6/PB8     |
| PD9           | 16   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PC8/PC7/PB9 |
| IOVDD/USBAVDD | 17   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PC9         |
| HUSBDM        | 18   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | VCM         |
| HUSBDP        | 19   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | AUVSS       |
| PD8           | 20   |         |         |       |     |           |     |     |     |     |     |         |     |      |       |       |         |         |         |         | PC10        |
|               |      | 21      | 22      | 23    | 24  | 25        | 26  | 27  | 28  | 29  | 30  | 31      | 32  | 33   | 34    | 35    | 36      | 37      | 38      | 39      | 40          |
|               |      | DVDD    | PF5     | PF4   | PF3 | FSPG/PC13 | PF2 | PF1 | PF0 | PR1 | PR0 | RTCVD33 | PV1 | PV0  | IOVDD | DCVDD | VBAT    | SW      | PGND    | PC12    | PC11        |

Figure 2-1 AC7923B Pin Assignment

## 2.2 Pin Description

Table 2-2-1 AC7923B Pin Description

| Pin No. | Name    | Type | IO Initial State | Description                                                                        |
|---------|---------|------|------------------|------------------------------------------------------------------------------------|
| 1       | WVDD14  | P    | --               | Wireless 1.4V Power                                                                |
| 2       | XOSCI   | I    | --               | Crystal Oscillator Input                                                           |
| 3       | XOSCO   | O    | --               | Crystal Oscillator Output                                                          |
| 4       | PE13    | I/O  | Z                | --                                                                                 |
| 5       | PE12    | I/O  | Z                | --                                                                                 |
| 6       | PE11    | I/O  | Z                | --                                                                                 |
| 7       | PE10    | I/O  | Z                | --                                                                                 |
| 8       | IOVDD2  | P    | --               | IO Power for PE10~PE13                                                             |
| 9       | PE0     | I/O  | Z                | SD1_CLK(B)                                                                         |
| 10      | PD15    | I/O  | Z                | SD1_CMD(B)                                                                         |
| 11      | PD14    | I/O  | Z                | SD1_DATA0(B)                                                                       |
| 12      | PD13    | I/O  | Z                | SD1_DATA1(B)                                                                       |
| 13      | PD12    | I/O  | Z                | SD1_DATA2(B)                                                                       |
| 14      | PD11    | I/O  | Z                | SD1_DATA3(B)                                                                       |
| 15      | PD10    | I/O  | Z                | ADC6(ADC Input Channel 6)<br>IO Wakeup Channel 6                                   |
| 16      | PD9     | I/O  | Z                | ADC5(ADC Input Channel 5)<br>IO Wakeup Channel 5                                   |
| 17      | USBAVDD | P    | --               | High Speed USB Power                                                               |
|         | IOVDD   | P    | --               | IO Power for PA1~PA15, PB0~PB9, PC6~PC13, PD8~PD15, PE0, PF0~PF5, PV0~PV1          |
| 18      | HUSBDM  | I/O  | 15kΩ Pull-down   | High Speed USB Negative Data                                                       |
| 19      | HUSBDP  | I/O  | 15kΩ Pull-down   | High Speed USB Positive Data                                                       |
| 20      | PD8     | I/O  | 10kΩ Pull-up     | MCLR(Device Reset)<br>ADC4(ADC Input Channel 4)<br>SD Power<br>IO Wakeup Channel 4 |
| 21      | DVDD    | P    | --               | Digital Logic Power                                                                |
| 22      | PF5     | I/O  | Z                | SFCTZ_DO<br>SPITZ_DO                                                               |
| 23      | PF4     | I/O  | Z                | SFCTZ_CLK<br>SPITZ_CLK                                                             |
| 24      | PF3     | I/O  | Z                | SFCTZ_DATA3<br>SPITZ_DATA3                                                         |
| 25      | FSPG    | I/O  | Z                | Flash Power Output                                                                 |
|         | PC13    | I/O  | Z                | ADC13(ADC Input Channel 13)<br>IO Wakeup Channel 13                                |



| Pin No. | Name     | Type | IO Initial State | Description                                                             |
|---------|----------|------|------------------|-------------------------------------------------------------------------|
| 26      | PF2      | I/O  | Z                | SFCTZ_DATA2<br>SPITZ_DATA2                                              |
| 27      | PF1      | I/O  | Z                | SFCTZ_DI<br>SPITZ_DI                                                    |
| 28      | PF0      | I/O  | Z                | SFCTZ_CS<br>SPITZ_CS                                                    |
| 29      | PR1      | I/O  | Z                | 32k Crystal Oscillator Output                                           |
| 30      | PR0      | I/O  | Z                | 32k Crystal Oscillator Input                                            |
| 31      | RTCVDD33 | P    | --               | RTC Power for PR0~PR1                                                   |
| 32      | PV1      | I/O  | Z                | AVDD18                                                                  |
| 33      | PV0      | I/O  | Z                | AVDD28                                                                  |
| 34      | IOVDD    | P    | --               | IO Power                                                                |
| 35      | DCVDD    | P    | --               | DCDC Power                                                              |
| 36      | VBAT     | P    | --               | Battery Input                                                           |
| 37      | SW       | P    | --               | Buck DCDC Switch Port                                                   |
| 38      | PGND     | G    | --               | Ground of Buck DC-DC converter                                          |
| 39      | PC12     | I/O  | Z                | AIN_BN0(Audio ADC Negative Input)                                       |
| 40      | PC11     | I/O  | Z                | AIN_BP0(Audio ADC Positive Input)                                       |
| 41      | PC10     | I/O  | 10kΩ Pull-down   | LVD(External Low Voltage Detection Input)<br>MICBIASB (MIC Bias Output) |
| 42      | AUVSS    | G    | --               | Audio Ground                                                            |
| 43      | VCM      | P    | --               | Audio Reference Power                                                   |
| 44      | PC9      | I/O  | Z                | MICBIASA (MIC Bias Output)<br>Right Channel DAC Output                  |
| 45      | PC8      | I/O  | Z                | Left Channel DAC Output                                                 |
|         | PC7      | I/O  | Z                | AIN_AP0(Audio ADC Positive Input)                                       |
|         | PB9      | I/O  | Z                | LCD_DATA17(A/B)                                                         |
| 46      | PC6      | I/O  | Z                | AIN_AN0(Audio ADC Negative Input)                                       |
|         | PB8      | I/O  | Z                | LCD_DATA16(A/B)                                                         |
| 47      | PB7      | I/O  | Z                | LCD_DATA15(A/B)<br>Sensor1_D7(A)<br>PAP_D15(A/B)<br>SD0_CLK(D)          |
| 48      | PB6      | I/O  | Z                | LCD_DATA14(A/B)<br>Sensor1_D6(A)<br>PAP_D14(A/B)<br>SD0_CMD(D)          |

| Pin No. | Name    | Type | IO Initial State | Description                                                      |
|---------|---------|------|------------------|------------------------------------------------------------------|
| 49      | PB5     | I/O  | Z                | LCD_DATA13(A/B)<br>Sensor1_D5(A)<br>PAP_D13(A/B)<br>SD0_DATA0(D) |
| 50      | PB4     | I/O  | Z                | LCD_DATA12(A/B)<br>Sensor1_D4(A)<br>PAP_D12(A/B)<br>SD0_DATA1(D) |
| 51      | PB3     | I/O  | Z                | LCD_DATA11(A/B)<br>Sensor1_D3(A)<br>PAP_D11(A/B)<br>SD0_DATA2(D) |
| 52      | PB2     | I/O  | Z                | LCD_DATA10(A/B)<br>Sensor1_D2(A)<br>PAP_D10(A/B)<br>SD0_DATA3(D) |
| 53      | PB1     | I/O  | Z                | LCD_DATA9(A/B)<br>Sensor1_D1(A)<br>PAP_D9(A/B)                   |
| 54      | PB0     | I/O  | Z                | LCD_DATA8(A/B)<br>Sensor1_D0(A)<br>PAP_D8(A/B)                   |
| 55      | PA15    | I/O  | Z                | LCD_SYNC2(A/B)<br>Sensor1_SYNC1(A)                               |
| 56      | PA14    | I/O  | Z                | LCD_SYNC1(A/B)<br>Sensor1_SYNC0(A)<br>PAP_RD(A/B)                |
| 57      | PA13    | I/O  | Z                | LCD_DCLK(A/B)<br>Sensor1_CLK(A)                                  |
| 58      | PA12    | I/O  | Z                | LCD_SYNC0(A/B)<br>PAP_WR(A/B)                                    |
| 59      | DVDD    | P    | --               | Digital Logic Power                                              |
| 60      | IOVDD   | P    | --               | IO Power                                                         |
|         | AVD33   | P    | --               | Analog 3.3V Power                                                |
| 61      | CSI_D1P | I    | --               | MIPI CSI D1P                                                     |
| 62      | CSI_D1N | I    | --               | MIPI CSI D1N                                                     |
| 63      | CSI_D0P | I    | --               | MIPI CSI D0P                                                     |
| 64      | CSI_D0N | I    | --               | MIPI CSI D0N                                                     |
| 65      | AVD12   | P    | --               | Analog 1.2V Power                                                |

| Pin No. | Name   | Type | IO Initial State | Description                                                              |
|---------|--------|------|------------------|--------------------------------------------------------------------------|
| 66      | PA11   | I/O  | Z                | LCD_DATA7(A)<br>Sensor0_D9(B)<br>PAP_D7(A)                               |
| 67      | PA10   | I/O  | Z                | SFC1_DO<br>LCD_DATA6(A)<br>Sensor0_D8(B)<br>PAP_D6(A)<br>SD0_DATA3(B)    |
| 68      | PA9    | I/O  | Z                | SFC1_CLK<br>LCD_DATA5(A)<br>Sensor0_D7(B)<br>PAP_D5(A)<br>SD0_DATA2(B)   |
| 69      | PA8    | I/O  | Z                | SFC1_DATA3<br>LCD_DATA4(A)<br>Sensor0_D6(B)<br>PAP_D4(A)<br>SD0_DATA1(B) |
| 70      | PA7    | I/O  | Z                | LCD_DATA3(A)<br>Sensor0_D5(B)<br>PAP_D3(A)<br>SD0_CLK(B)                 |
| 71      | PA6    | I/O  | Z                | SFC1_DATA2<br>LCD_DATA2(A)<br>Sensor0_D4(B)<br>PAP_D2(A)<br>SD0_CMD(B)   |
| 72      | PA5    | I/O  | Z                | SFC1_DI<br>LCD_DATA1(A)<br>Sensor0_D3(B)<br>PAP_D1(A)<br>SD0_DATA0(B)    |
| 73      | PA4    | I/O  | Z                | SFC1_CS<br>LCD_DATA0(A)<br>Sensor0_D2(B)<br>PAP_D0(A)                    |
| 74      | PA3    | I/O  | Z                | Sensor0_CLK(B)                                                           |
| 75      | PA2    | I/O  | Z                | Sensor0_SYNC1(B)                                                         |
| 76      | PA1    | I/O  | Z                | Sensor0_SYNC0(B)                                                         |
| 77      | WVSS   | G    | --               | Ground of Wireless                                                       |
| 78      | WVDD14 | P    | --               | Wireless 1.4V Power                                                      |

| Pin No. | Name   | Type | IO Initial State | Description           |
|---------|--------|------|------------------|-----------------------|
| 79      | WVDD33 | P    | --               | Wireless 3.3V Power   |
| 80      | ANTP   | RF   | --               | Antenna Positive Port |

**Note**

1. IO initial state abbreviations Z--High resistance, H--High level, L--Low level, X--May be changed during power on.
2. Timer, CAN, MCPWM, QDEC, UART, LEDC, I<sup>2</sup>C, I<sup>2</sup>S and SPI functions can be remapped to any I/O (except PF/PR/PV/CSI).

**Table 2-2-2 Pin Types Description**

| Pin Type | Description | Pin Type | Description     |
|----------|-------------|----------|-----------------|
| P        | Power       | I/O      | Input or Output |
| G        | Ground      | I        | Input           |
| RF       | RF antenna  | O        | Output          |

## 3 Electrical Characteristics

### 3.1 Absolute Maximum Ratings

Table 3-1 Absolute Maximum Ratings

| Symbol   | Parameter             | Min  | Max  | Unit |
|----------|-----------------------|------|------|------|
| Topt     | Operating temperature | -20  | +85  | °C   |
| Tstg     | Storage temperature   | -65  | +150 | °C   |
| VBAT     | Supply Voltage        | -0.3 | 5.5  | V    |
| IOVDD    |                       | -0.3 | 3.6  | V    |
| IOVDD2   |                       | -0.3 | 3.6  | V    |
| RTCVDD33 |                       | -0.3 | 3.6  | V    |
| DCVDD    |                       | -0.3 | 1.54 | V    |
| WVDD33   |                       | -0.3 | 3.6  | V    |
| WVDD14   |                       | -0.3 | 1.54 | V    |
| USBAVDD  |                       | -0.3 | 3.6  | V    |
| AVD33    |                       | -0.3 | 3.6  | V    |
| AVD12    |                       | -0.3 | 1.54 | V    |
| GPIO     | Input voltage of GPIO | -0.3 | 3.6  | V    |

#### Note

1. Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device.

### 3.2 ESD Ratings

Table 3-2 ESD Ratings

| Parameter           | Typ   | Test pin | Reference standard          |
|---------------------|-------|----------|-----------------------------|
| Human Body Mode     | ±4kV  | All pins | JEDEC EIA/JESD22-A114       |
| Machine Mode        | ±300V | All pins | JEDEC EIA/JESD22-A115       |
| Charge Device Model | ±1kV  | All pins | ANSI/ESDA/JEDEC JS-002-2022 |

### 3.3 PMU Characteristics

Table 3-3 PMU Characteristics

| Symbol         | Parameter       | Conditions               | Min | Typ | Max | Unit |
|----------------|-----------------|--------------------------|-----|-----|-----|------|
| VBAT           | Power supply    | --                       | 2.7 | 3.7 | 5.5 | V    |
| Operating mode |                 |                          |     |     |     |      |
| Symbol         | Parameter       | Conditions               | Min | Typ | Max | Unit |
| IOVDD          | Voltage output  | --                       | 2.4 | 3.3 | 3.4 | V    |
|                | Loading current | IOVDD=3.3V@VBAT = 3.9V   | --  | --  | 200 | mA   |
| AVDD28         | Voltage output  | --                       | 2.5 | 2.8 | 3.2 | V    |
|                | Loading current | AVDD28=2.8V@IOVDD = 3.3V | --  | --  | 100 | mA   |
| AVDD18         | Voltage output  | --                       | 1.5 | 1.8 | 2.2 | V    |

| Symbol         | Parameter       | Conditions                        | Min | Typ | Max | Unit |
|----------------|-----------------|-----------------------------------|-----|-----|-----|------|
|                | Loading current | AVDD18=1.8V@IOVDD = 3.3V          | --  | --  | 60  | mA   |
| DCVDD          | Voltage output  | --                                | --  | 1.4 | --  | V    |
|                | Loading current | DCVDD=1.4V@IOVDD = 3.3V, LDO mode | --  | --  | 60  | mA   |
|                |                 | DCVDD=1.4V@VBAT = 3.7V, DCDC mode | --  | --  | 180 | mA   |
| Low Power mode |                 |                                   |     |     |     |      |
| Symbol         | Parameter       | Conditions                        | Min | Typ | Max | Unit |
| IOVDD          | Loading current | IOVDD=3.0V@VBAT = 3.7V            | --  | --  | 10  | mA   |

### 3.4 IO Characteristics

Table 3-4 IO Characteristics

| Input Characteristics  |                          |                                                                        |                                                                                                       |                                                |     |      |
|------------------------|--------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------|-----|------|
| Symbol                 | Parameter                | Conditions                                                             | IO                                                                                                    | Min                                            | Max | Unit |
| V <sub>IL</sub>        | Low-Level Input Voltage  | IOVDD2 = 3.0V                                                          | PE10~PE13                                                                                             | -0.3                                           | 1.4 | V    |
|                        |                          | IOVDD2 = 1.8V                                                          | PE10~PE13                                                                                             | -0.3                                           | 0.7 | V    |
|                        |                          | IOVDD = 3.0V                                                           | PA1~PA15<br>PB0~PB7<br>PC6~PC13<br>PD8~PD15<br>PE0, PF0~PF5<br>HUSBDP<br>HUSBDM<br>PR0~PR1<br>PV0~PV1 | -0.3                                           | 1.4 | V    |
| V <sub>IH</sub>        | High-Level Input Voltage | IOVDD2 = 3.0V                                                          | PE10~PE13                                                                                             | 1.7                                            | 3.3 | V    |
|                        |                          | IOVDD2 = 1.8V                                                          | PE10~PE13                                                                                             | 1.1                                            | 2.0 | V    |
|                        |                          | IOVDD = 3.0V                                                           | PA1~PA15<br>PB0~PB7<br>PC6~PC13<br>PD8~PD15<br>PE0, PF0~PF5<br>HUSBDP<br>HUSBDM<br>PR0~PR1<br>PV0~PV1 | 1.7                                            | 3.3 | V    |
| Output Characteristics |                          |                                                                        |                                                                                                       |                                                |     |      |
| Symbol                 | Parameter                | Conditions                                                             | IO                                                                                                    | Typ                                            |     | Unit |
| I <sub>OL</sub>        | Output Current           | IOVDD2 = 3.0V<br>Voutput = 0.3V<br><br>IOVDD2 = 1.8V<br>Voutput = 0.2V | PE10~PE13                                                                                             | 2.5(HD=0)<br>8(HD=1)<br>18.5(HD=2)<br>24(HD=3) |     | mA   |

|                                     |                      |                                    |                                                                          |                                                |      |
|-------------------------------------|----------------------|------------------------------------|--------------------------------------------------------------------------|------------------------------------------------|------|
|                                     |                      | IOVDD = 3.0V<br>Voutput = 0.3V     | PA1~PA15<br>PB0~PB7<br>PC6~PC13<br>PD8~PD15<br>PE0, PF0~PF5              | 2.5(HD=0)<br>8(HD=1)<br>18.5(HD=2)<br>24(HD=3) | mA   |
|                                     |                      |                                    | PR0~PR1<br>PV0~PV1                                                       | 2.5(HD=0)<br>18.5(HD=1)                        | mA   |
|                                     |                      |                                    | HUSBDP<br>HUSBDM                                                         | 8                                              | mA   |
| I <sub>OH</sub>                     | Output Current       | IOVDD2 = 3.0V<br>Voutput = 2.7V    | PE10~PE13                                                                | 2.5(HD=0)<br>8(HD=1)<br>18.5(HD=2)<br>24(HD=3) | mA   |
|                                     |                      | IOVDD2 = 1.8V<br>Voutput = 1.6V    |                                                                          |                                                |      |
|                                     |                      | IOVDD = 3.0V<br>Voutput = 2.7V     | PA1~PA15<br>PB0~PB7<br>PC6~PC13<br>PD8~PD15<br>PE0, PF0~PF5              | 2.5(HD=0)<br>8(HD=1)<br>18.5(HD=2)<br>24(HD=3) | mA   |
|                                     |                      |                                    | PR0~PR1<br>PV0~PV1                                                       | 2.5(HD=0)<br>18.5(HD=1)                        | mA   |
|                                     |                      |                                    | HUSBDP<br>HUSBDM                                                         | 8                                              | mA   |
|                                     |                      |                                    |                                                                          |                                                |      |
| Internal Resistance Characteristics |                      |                                    |                                                                          |                                                |      |
| Symbol                              | Parameter            | Conditions                         | IO                                                                       | Typ                                            | Unit |
| R <sub>pu</sub>                     | Pull-up Resistance   | IOVDD = 3.0V<br>IOVDD2 = 3.0V/1.8V | PA1~PA15<br>PB0~PB7<br>PC6~PC13<br>PD8~PD15<br>PE0, PE10~PE13<br>PF0~PF5 | 10k                                            | Ω    |
|                                     |                      |                                    | HUSBDP                                                                   | 1.5k(PU=1)<br>1k(PU=2/3)                       | Ω    |
| R <sub>pd</sub>                     | Pull-down Resistance | IOVDD = 3.0V<br>IOVDD2 = 3.0V/1.8V | PA1~PA15<br>PB0~PB7<br>PC6~PC13<br>PD8~PD15<br>PE0, PE10~PE13<br>PF0~PF5 | 10k                                            | Ω    |
|                                     |                      |                                    | HUSBDP<br>HUSBDM                                                         | 15k                                            | Ω    |

**Note**

1.Internal pull-up/pull-down resistance accuracy ±20%.

### 3.5 Audio DAC Characteristics

Table 3-5 Stereo DAC Characteristics

| Parameter          | Conditions                                                                                    | Min | Typ | Max | Unit  |
|--------------------|-----------------------------------------------------------------------------------------------|-----|-----|-----|-------|
| Resolution         | --                                                                                            | --  | 16  | --  | bits  |
| Output Sample Rate | --                                                                                            | 8   | --  | 96  | kHz   |
| SNR                | Differential Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ   | --  | 103 | --  | dB    |
|                    | Single Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ         | --  | 100 | --  | dB    |
| Dynamic Range      | Differential Mode<br>Fin=1kHz@-60dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ | --  | 103 | --  | dB    |
|                    | Single Mode<br>Fin=1kHz@-60dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ       | --  | 100 | --  | dB    |
| THD+N              | Differential Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ   | --  | -87 | --  | dB    |
|                    | Single Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ         | --  | -75 | --  | dB    |
| Noise Floor        | Differential Mode<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ                                   | --  | 6.4 | --  | uVrms |
|                    | Single Mode<br>B/W=20Hz~20kHz A-Weighted<br>load=10kΩ                                         | --  | 5.4 | --  | uVrms |



### 3.6 Audio ADC Characteristics

Table 3-6 Audio ADC Characteristics

| Parameter         | Conditions                                                                                             | Min | Typ  | Max | Unit |
|-------------------|--------------------------------------------------------------------------------------------------------|-----|------|-----|------|
| Resolution        | --                                                                                                     | --  | 16   | --  | bits |
| Input Sample Rate | --                                                                                                     | 8   | --   | 48  | kHz  |
| SNR               | Differential input Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>ADC gain=0dB   | --  | 95   | --  | dB   |
|                   | Single-ended input Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>ADC gain=0dB   | --  | 92   | --  | dB   |
| Dynamic Range     | Differential input Mode<br>Fin=1kHz@-60dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>ADC gain=0dB | --  | 95   | --  | dB   |
|                   | Single-ended input Mode<br>Fin=1kHz@-60dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>ADC gain=0dB | --  | 92   | --  | dB   |
| THD+N             | Differential input Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>ADC gain=0dB   | --  | -87  | --  | dB   |
|                   | Single-ended input Mode<br>Fin=1kHz@0dBFS<br>Fs=44.1kHz<br>B/W=20Hz~20kHz A-Weighted<br>ADC gain=0dB   | --  | -81  | --  | dB   |
| Analogue Gain     |                                                                                                        | -6  | --   | 28  | dB   |
| Max Input Level   | Differential input Mode<br>ADC gain=0dB                                                                | --  | 0.7  | --  | Vrms |
|                   | Single-ended input Mode<br>ADC gain=0dB                                                                | --  | 0.35 | --  | Vrms |

### 3.7 BT Characteristics

#### 3.7.1 Transmitter

Table 3-7-1 Transmitter characteristics

| Parameter                 | Conditions                     | Min | Typ | Max | Unit |
|---------------------------|--------------------------------|-----|-----|-----|------|
| Maximum RF Transmit Power | BR                             | --  | 19  | --  | dBm  |
| Maximum RF Transmit Power | EDR $\pi/4$ DQPSK<br>EDR 8DPSK | --  | 19  | --  | dBm  |
| Relative Transmit Power   | EDR $\pi/4$ DQPSK<br>EDR 8DPSK | --  | 1.5 | --  | dB   |
| Maximum RF Transmit Power | BLE-1Mbps                      | --  | 19  | --  | dBm  |

#### 3.7.2 Receiver

Table 3-7-2 Receiver characteristics

| Parameter   | Conditions        | Min | Typ   | Max | Unit |
|-------------|-------------------|-----|-------|-----|------|
| Sensitivity | BR                | --  | -95.5 | --  | dBm  |
|             | EDR $\pi/4$ DQPSK | --  | -96   | --  | dBm  |
|             | EDR 8DPSK         | --  | -88   | --  | dBm  |
|             | BLE-1Mbps         | --  | -98   | --  | dBm  |
|             | BLE-2Mbps         | --  | -95   | --  | dBm  |
|             | BLE-S2            | --  | -101  | --  | dBm  |
|             | BLE-S8            | --  | -106  | --  | dBm  |

### 3.8 WiFi Characteristics

#### 3.8.1 Transmitter

TX Power with Spectral Mask and EVM Meeting 802.11 Standards.

Table 3-8-1 Transmitter characteristics

| Parameter | Conditions             | Min | Typ | Max | Unit |
|-----------|------------------------|-----|-----|-----|------|
| TX Power  | 802.11b, 1 Mbps, DSSS  | --  | 19  | --  | dBm  |
|           | 802.11b, 11 Mbps, CCK  | --  | 19  | --  | dBm  |
|           | 802.11g, 6 Mbps, OFDM  | --  | 17  | --  | dBm  |
|           | 802.11g, 54 Mbps, OFDM | --  | 14  | --  | dBm  |
|           | 802.11n, HT20, MCS0    | --  | 17  | --  | dBm  |
|           | 802.11n, HT20, MCS7    | --  | 13  | --  | dBm  |
|           | 802.11n, HT40, MCS0    | --  | 17  | --  | dBm  |
|           | 802.11n, HT40, MCS7    | --  | 12  | --  | dBm  |

### 3.8.2 Receiver

For RX tests, the PER (packet error rate) limit is 8% for 802.11b, and 10% for 802.11g/n.

**Table 3-8-2 Receiver characteristics**

| Parameter   | Conditions             | Min | Typ   | Max | Unit |
|-------------|------------------------|-----|-------|-----|------|
| Sensitivity | 802.11b, 1 Mbps, DSSS  | --  | -97   | --  | dBm  |
|             | 802.11b, 2 Mbps, DSSS  | --  | -94.5 | --  | dBm  |
|             | 802.11b, 5.5 Mbps, CCK | --  | -93   | --  | dBm  |
|             | 802.11b, 11 Mbps, CCK  | --  | -90   | --  | dBm  |
|             | 802.11g, 6 Mbps, OFDM  | --  | -93   | --  | dBm  |
|             | 802.11g, 9 Mbps, OFDM  | --  | -92   | --  | dBm  |
|             | 802.11g, 12 Mbps, OFDM | --  | -91   | --  | dBm  |
|             | 802.11g, 18 Mbps, OFDM | --  | -89   | --  | dBm  |
|             | 802.11g, 24 Mbps, OFDM | --  | -86   | --  | dBm  |
|             | 802.11g, 36 Mbps, OFDM | --  | -83   | --  | dBm  |
|             | 802.11g, 48 Mbps, OFDM | --  | -79   | --  | dBm  |
|             | 802.11g, 54 Mbps, OFDM | --  | -77   | --  | dBm  |
|             | 802.11n, HT20, MCS0    | --  | -93   | --  | dBm  |
|             | 802.11n, HT20, MCS1    | --  | -90.5 | --  | dBm  |
|             | 802.11n, HT20, MCS2    | --  | -88   | --  | dBm  |
|             | 802.11n, HT20, MCS3    | --  | -84.5 | --  | dBm  |
|             | 802.11n, HT20, MCS4    | --  | -81.5 | --  | dBm  |
|             | 802.11n, HT20, MCS5    | --  | -77   | --  | dBm  |
|             | 802.11n, HT20, MCS6    | --  | -75   | --  | dBm  |
|             | 802.11n, HT20, MCS7    | --  | -74   | --  | dBm  |
|             | 802.11n, HT40, MCS0    | --  | -89   | --  | dBm  |
|             | 802.11n, HT40, MCS1    | --  | -86   | --  | dBm  |
|             | 802.11n, HT40, MCS2    | --  | -84   | --  | dBm  |
|             | 802.11n, HT40, MCS3    | --  | -80   | --  | dBm  |
|             | 802.11n, HT40, MCS4    | --  | -77.5 | --  | dBm  |
|             | 802.11n, HT40, MCS5    | --  | -72.5 | --  | dBm  |
|             | 802.11n, HT40, MCS6    | --  | -71.5 | --  | dBm  |
|             | 802.11n, HT40, MCS7    | --  | -70   | --  | dBm  |
|             | 802.11n, HT40, MCS32   | --  | -89   | --  | dBm  |

## 4 Package Information

### 4.1 QFN80\_8\*8mm

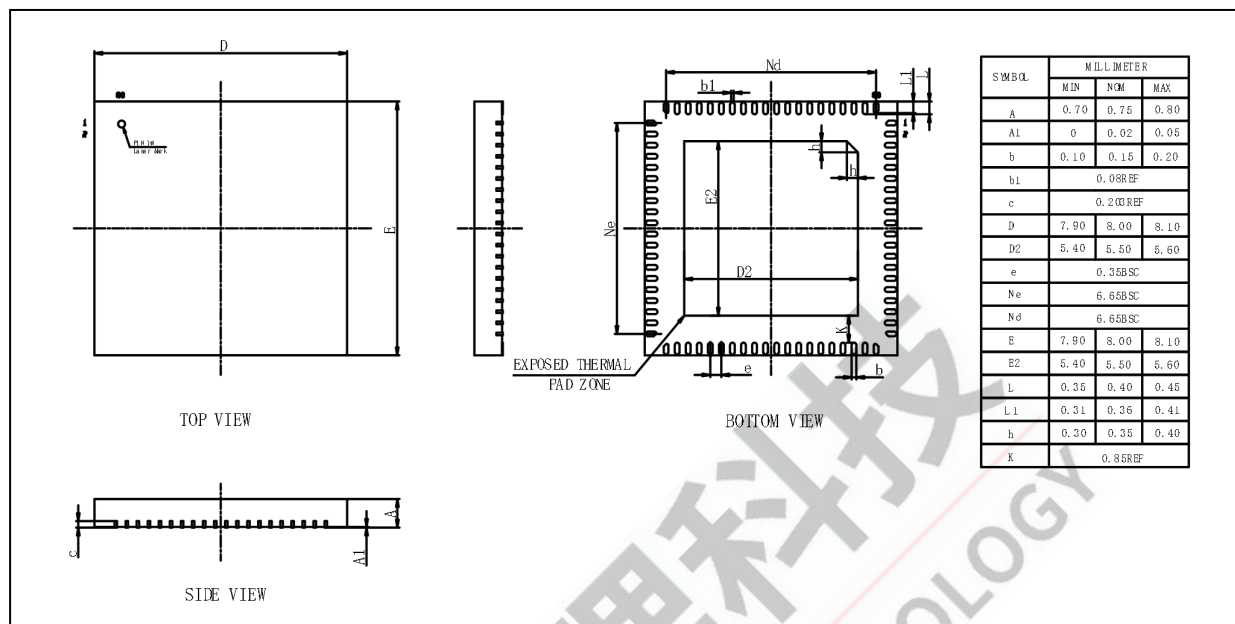


Figure 4-1 AC7923B Package

## 5 IC Marking Information

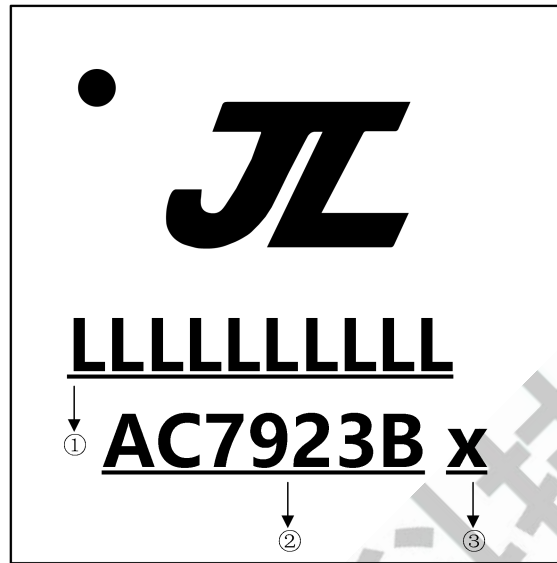


Figure 5-1 AC7923B Package Outline

- ① Production Batch
- ② Chip Model
- ③ Built-in DDR size
  - 0 No Flash Memory
  - 2 2Mbit flash
  - 4 4Mbit flash
  - 8 8Mbit flash
  - 6 16Mbit flash
  - 3 32Mbit flash
  - 5 64Mbit flash
  - 7 128Mbit flash
  - A 1Mx16 SDRAM
  - B 4Mx16 SDRAM
  - E 4Mx16bit DDR1
  - F 8Mx16bit DDR1
  - G 16Mx16bit DDR1

## 6 Solder-Reflow Condition

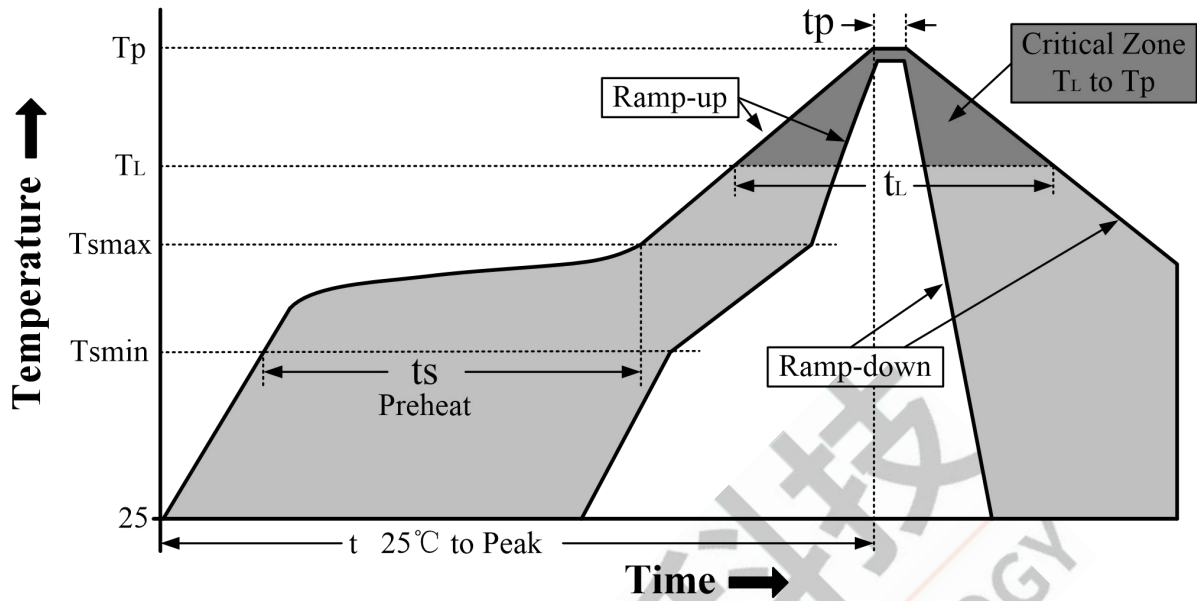


Figure 6-1 Classification Reflow Profile

Table 6-1 Classification Profiles

| Profile Feature                                                   |                                                  | Sn-Pb Eutectic Assembly | Pb-Free Assembly |
|-------------------------------------------------------------------|--------------------------------------------------|-------------------------|------------------|
| Preheat/Soak                                                      | Temperature Min ( $T_{smin}$ )                   | 100°C                   | 150°C            |
|                                                                   | Temperature Max ( $T_{smax}$ )                   | 150°C                   | 200°C            |
|                                                                   | Time ( $t_s$ ) from ( $T_{smin}$ to $T_{smax}$ ) | 60-120 seconds          | 60-180 seconds   |
| Average ramp-up rate ( $T_{smax}$ to $T_p$ )                      |                                                  | 3°C/second max          | 3°C/second max   |
| Liquidous temperature ( $T_L$ )                                   |                                                  | 183°C                   | 217°C            |
| Time ( $t_L$ ) maintained above $T_L$                             |                                                  | 60-150 seconds          | 60-150 seconds   |
| Peak package body temperature ( $T_p$ )                           |                                                  | See Table 6-2           | See Table 6-3    |
| Time within 5°C of actual Peak Temperature ( $t_p$ ) <sup>2</sup> |                                                  | 10-30 seconds           | 20-40 seconds    |
| Ramp-down rate ( $T_p$ to $T_L$ )                                 |                                                  | 6°C/second max          | 6°C/second max   |
| Time 25°C to peak temperature                                     |                                                  | 6 minutes max           | 8 minutes max    |

**Note**

- 1.All temperatures refer to topside of the package, measured on the package body surface
- 2.Time within 5°C of actual peak temperature ( $t_p$ ) specified for the reflow profiles is a "supplier" and "user" maximum.

Table 6-2 SnPb Classification Temperature

| Package Thickness | Volume mm <sup>3</sup><br>< 350 | Volume mm <sup>3</sup><br>≥ 350 |
|-------------------|---------------------------------|---------------------------------|
| <2.5 mm           | 240 +0/-5°C                     | 225 +0/-5°C                     |
| ≥2.5 mm           | 225 +0/-5°C                     | 225 +0/-5°C                     |

**Table 6-3 Pb-free - Classification Temperature**

| Package Thickness | Volume mm <sup>3</sup><br>< 350 | Volume mm <sup>3</sup><br>350 - 2000 | Volume mm <sup>3</sup><br>> 2000 |
|-------------------|---------------------------------|--------------------------------------|----------------------------------|
| < 1.6mm           | 260°C                           | 260°C                                | 260°C                            |
| 1.6 mm - 2.5mm    | 260°C                           | 250°C                                | 245°C                            |
| > 2.5mm           | 250°C                           | 245°C                                | 245°C                            |

**Note**

1.\*Tolerance The device manufacturer/supplier shall assure process compatibility up to and including the stated classification temperature (this means Peak reflow temperature +0°C.For example 260°C+0°C)at the rated MSL level.